

Bhargi Bhalara

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SUMMARY

Dynamic hardware engineer with over eight years of experience specializing in FPGA-based solutions and high-speed digital design. Proficient in the design and development of Ethernet switching platforms, showcasing expertise in modular chassis-based products, terabit switch fabrics, and embedded CPU subsystems. Proven track record of enhancing system architecture and automating test processes, leading to significant improvements in efficiency and accuracy throughout the product lifecycle.

EXPERIENCE

Electrical Systems Engineer

Reflexion Medical, Inc.

Jun 2021 - May 2025

- Implemented end-to-end FPGA design flow including Synthesis, Implementation, Place and Route, and Bitstream Generation for Xilinx Ultrascale FPGAs using Vivado design suite, improving design cycle efficiency by 30%.
- Optimized the existing state machine, achieving a reduction in operational delay by 3-4 clock cycles, which enhanced overall system performance by 20%.
- Automated regression test cases using Python and standard verification methodologies, resulting in a 50% reduction in testing time and increased accuracy of results.
- Developed embedded test programs and prepared validation protocols for bare metal testing, leading to a 40% improvement in testing coverage and reliability.
- Debugged High-Speed Serial Interfaces like PCIe and CAN Bus using protocol analyzers, ensuring compliance with industry standards and reducing error rates by 15%.
- Initiated engineering change orders and supported the design transfer process for updated FPGA designs, facilitating a seamless transition to production.

Electrical Engineer

Mirion Technologies (Capintec) Inc.

Apr 2017 - Jun 2021

- Remodeled RTL design logic for FPGA/CPLD used in Multi-Channel Analyzers, enhancing design efficiency and reducing time to market by 25%.
- Supported design changes for electronic components like Microcontroller, SDRAM, PMIC, and similar CMOS devices in PCBs, ensuring compliance with ISO-13485 and IEC-60601 standards.
- Delivered design changes to production efficiently, adhering to regulatory requirements and enhancing product reliability by 30%.
- Led hardware validation for Dose Calibrators, Thyroid Uptake Systems, and Radiation Measurement Devices, ensuring measurement accuracy and reliability, which resulted in a 20% increase in customer satisfaction.

Electronics Engineer

Reliance Electronics Inc.

Jan 2017 - Sep 2017

- Designed and validated printed circuit boards (PCB) using custom test fixtures, ensuring correct functionality post-manufacture and reducing error rates by 10%.
- Structured the BOM for circuit designs by selecting appropriate parts and updating designs due to EOL/obsolescence, ensuring continuity in production and minimizing delays, which improved production timelines by 15%.

Design Engineer

Scalable Systems Research Labs

Feb 2016 - Dec 2016

- Structured IEEE Floating Point Numbers Multiplier in Verilog for the Floating Point Unit used in RISC-V CPU, enhancing computational performance by 25%.
- Reduced delay on carry propagation by implementing vertical slice compression technique, therefore increasing the speed of the processor by 15%.

FPGA Intern

Finisar Corporation

Jun 2016 - Sep 2016

- Developed comprehensive Verilog code for 10G Ethernet transceivers on Xilinx Kintex 7 FPGA, contributing to a product that improved data transmission speeds by 40%.
- Enhanced power efficiency by optimizing the state machine for Microsemi FPGA, resulting in a 30% reduction in power consumption.

Instructional Student Assistant

San Jose State University

Aug 2014 - Dec 2014

- Assisted students with their Advance Digital Logic Course, teaching digital design concepts like combinational and sequential circuits, Logic Reduction Techniques, and simulation of VHDL code in ModelSim, which improved student performance by 20%.

EDUCATION

Masters of Science in Electrical Engineering

San Jose State University, San Jose

Dec 2015

GPA: 3.4

Bachelor of Engineering

Sarvajani College of Engineering and Technology

May 2013

SKILLS

AMBA (AHB, APB, ASB, AXI), ASIC, C, C++, Chipscope/ModelSim/Questasim, JIRA, JTAG, Lab Equipment and tools, Linux, Mentor Graphics PADS, Shell, SOC, System Verilog, USB, Verilog, CPU Architecture, GIT, HDMI, Intel Quartus II/Prime, Jenkins, Microchip Libero SoC, PCIe/PCI, Python, QNX, QuestaSim, RTOS, VHDL, Vitis HLS, VUnit verification framework, Xilinx ISE Design Suite, Xilinx Vivado Design Suite